

Lab 1: Film Ranking (The Basics)

```
Query    Query History
1  WITH RankedFilms AS (
2      SELECT c.name AS category, f.title AS film_title, f.length AS film_len,
3      DENSE_RANK() OVER (PARTITION BY c.name ORDER BY f.length DESC, f.title) AS rn
4      FROM film f
5      JOIN film_category fc ON f.film_id = fc.film_id
6      JOIN CATEGORY C ON c.category_id = fc.category_id
7  )
8  SELECT * FROM RankedFilms WHERE rn <= 3
```

Data Output Messages Notifications

Showing rows: 1 to 48 Page No: 1 of 1

	category character varying (25)	film_title character varying (255)	film_len smallint	rn bigint
1	Action	Darn Forrester	185	1
2	Action	Worst Banger	185	2
3	Action	Casualties Encino	179	3
4	Animation	Gangs Pride	185	1
5	Animation	Pond Seattle	185	2
6	Animation	Sons Interview	184	3
7	Children	Fury Murder	178	1
8	Children	Wrong Behavior	178	2
9	Children	Empire Malkovich	177	3
10	Classics	Conspiracy Spirit	184	1
11	Classics	Dracula Crystal	176	2
12	Classics	Lights Deer	174	3
13	Comedy	Control Anthem	185	1
14	Comedy	Saturn Name	182	2
15	Comedy	Searchers Wait	182	3

Total rows: 48 Query complete 00:00:00.037 LF Ln 8, Col 4

Lab 2: Growth Analysis (Time Series)

Query Query History

```

18 SELECT
19     month,
20     revenue,
21     LAG(revenue) OVER (
22         ORDER BY month
23     ) AS previous_revenue
24 FROM monthly_revenue
25 ORDER BY month;
26
27 WITH monthly_revenue AS (
28     SELECT
29         DATE_TRUNC('month', payment_date) AS month,
30         SUM(amount) AS revenue
31     FROM payment
32     GROUP BY DATE_TRUNC('month', payment_date)
33 )
34 SELECT
35     month,
36     revenue,
37     LAG(revenue) OVER (
38         ORDER BY month
39     ) AS previous_revenue,
40     (
41         (revenue - LAG(revenue) OVER (ORDER BY month))
42         / LAG(revenue) OVER (ORDER BY month)
43     ) * 100 AS growth_percent
44 FROM monthly_revenue
45 ORDER BY month;

```

Data Output Messages Notifications

Showing rows: 1 to 4 Page No: 1 of 1

	month timestamp without time zone	revenue numeric	previous_revenue numeric	growth_percent numeric
1	2007-02-01 00:00:00	8351.84	[null]	[null]
2	2007-03-01 00:00:00	23886.56	8351.84	186.0035632866530000
3	2007-04-01 00:00:00	28559.46	23886.56	19.56288389789069669300
4	2007-05-01 00:00:00	514.18	28559.46	-98.19961581906660700200

Total rows: 4 Query complete 00:00:00.039 LF Ln 45, Col 16

Lab 3: Customer Lifetime Value (Running Total)

```

47 SELECT
48     customer_id,
49     payment_date,
50     amount,
51     SUM(amount) OVER (
52         PARTITION BY customer_id
53         ORDER BY payment_date
54     ) AS cumulative_spend
55 FROM payment
56 ORDER BY customer_id, payment_date;

```

Data Output Messages Notifications

Showing rows: 1 to 1000 Page No: 1 of 15

	customer_id smallint	payment_date timestamp without time zone	amount numeric (5,2)	cumulative_spend numeric
1	1	2007-02-14 23:22:38.996577	5.99	5.99
2	1	2007-02-15 16:31:19.996577	0.99	6.98
3	1	2007-02-15 19:37:12.996577	9.99	16.97
4	1	2007-02-16 13:47:23.996577	4.99	21.96
5	1	2007-02-18 07:10:14.996577	4.99	26.95
6	1	2007-02-18 12:02:25.996577	0.99	27.94
7	1	2007-02-21 04:53:11.996577	3.99	31.93
8	1	2007-03-01 07:19:30.996577	4.99	36.92
9	1	2007-03-02 14:05:18.996577	3.99	40.91
10	1	2007-03-02 16:30:04.996577	0.99	41.90
11	1	2007-03-17 11:06:20.996577	4.99	46.89
12	1	2007-03-18 02:25:55.996577	0.99	47.88
13	1	2007-03-19 08:23:42.996577	0.99	48.87
14	1	2007-03-19 12:25:20.996577	2.99	51.86
15	1	2007-03-21 22:02:23.996577	0.99	52.85
16	1	2007-03-21 23:56:23.996577	1.99	54.84
17	1	2007-03-22 18:10:03.996577	2.99	57.83
18	1	2007-03-22 18:32:12.996577	5.99	63.82

Total rows: 14596 Query complete 00:00:00.047 LF Ln 73, Col 1

Lab 4: The "Top-N" Filter Challenge

Query

Query History

Explain

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WITH customer_spending AS (

SELECT

c.customer_id,

c.first_name,

c.last_name,

co.country,

SUM(p.amount) AS total_spend

FROM customer c

JOIN payment p

ON c.customer_id = p.customer_id

JOIN address a

ON c.address_id = a.address_id

JOIN city ci

ON a.city_id = ci.city_id

JOIN country co

Data Output

Messages

Notifications

Showing rows: 1 to 108

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	customer_id integer	first_name character varying (45)	last_name character varying (45)	country character varying (50)	total_spend numeric	spending_rank bigint
1	218	Vera	Mccoy	Afghanistan	67.82	1
2	176	June	Carroll	Algeria	151.68	1
3	320	Anthony	Schwab	American Samoa	47.85	1
4	528	Claude	Herzog	Angola	93.80	1
5	381	Bobby	Boudreau	Anguilla	99.68	1
6	560	Jordan	Archuleta	Argentina	129.71	1
7	41	Stephanie	Mitchell	Armenia	118.75	1
8	266	Nora	Herrera	Austria	113.74	1
9	334	Raymond	Mcwhorter	Azerbaijan	108.75	1
10	590	Seth	Hannon	Bahrain	108.76	1
11	21	Michelle	Clark	Bangladesh	146.68	1
12	144	Clara	Shaw	Belarus	189.60	1
13	431	Joel	Francisco	Bolivia	92.78	1
14	178	Marion	Snyder	Brazil	194.61	1
15	01	Lois	Butler	Brunei	107.66	1

Total rows: 108

Query complete 00:00:00.049

LF

Ln 93, Col 18